Zahib ordaring system

1. Problem Analysis and System Requirements

1.1 Problem Analysis

During peak hours, the university cafeteria faces the following challenges:

- Delays in receiving and delivering orders due to congestion at the counter
- Unclear order status when ready, increasing pressure on cafeteria staff

➢ **The objectives of designing a smart food ordering system:**

- Automated queue management
- Reduced waiting times
- Improved customer experience
- Simplified order management for staff

1.2 Functional Requirements

The system to be designed must include the following:

- Allowing customers to create an account and log in securely.
- Displaying a list of items available in the cafeteria.
- Enabling customers to add desired items and place orders.
- Storing order details in a database.
- Providing order tracking functionality.
- Allowing customers to rate their experience.
- Providing a control panel for employees to view and analyze orders.

1.3 Non-Functional Requirements

- Usability: Simple and intuitive interface
- Performance: Fast response time for user actions
- Security: Password hashing and secure data storage
- Reliability: Stable execution without crashes
- Maintainability: Well-structured and readable code

2. System Design Artefacts

2.1 UML Diagrams

1. **Use Case Diagram :**
   Actors: User, Staff

- User: Register, Login, Browse Menu, Place Order, Track Order, Rate Order
- Staff: Login, View Orders, Analyze Data

2. **Class Diagram :**

The system is structured into several logical components.

The User class represents the user entity, storing authentication details.

The Order class represents each order, including attributes such as food item, price, status, rating, and time.

The main system logic is handled through functional components responsible for operations such as login, registration, order processing, tracking, and analytics.

- User (username, password)
- Order (id, food, price, status, rating, time)
- System (handles logic and interaction)

3. **Entity Relationship Diagram (ERD)**

Entities:

- Users (username, password)
- Orders (id, username, food, price, status, rating, time)

Relationship:

- One user can have many orders (1-to-Many relationship)

3. Agile Methodology

The project was developed using an Agile approach, specifically inspired by the Scrum methodology.

The development process was divided into small iterations (sprints), where each sprint focused on implementing a specific set of features.

Initially, the team focused on core functionalities such as login, registration, and database setup. In later iterations, more advanced features were added, including order tracking, rating system, and data analytics.

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4. Key System Design

The system follows a modular design approach:

- User Interface Layer: Built using Tkinter
- Logic Layer: Handles operations such as login, ordering, and tracking
- Data Layer: SQLite database for storing users and orders

Key design decisions include:

- Separation of concerns between UI and database

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5. Implementation Decisions

- Python was chosen for its simplicity and readability
- Tkinter was used for GUI development due to its integration with Python
- SQLite was selected as a lightweight database solution
- Passwords were hashed using SHA-256 for security

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6. Project Progress Update

1. User Interface (UI)

- Designed the login interface
- Added a registration (Register) feature for new users
- Designed the menu using card layout (images + prices)
- Improved overall user experience (colors, layout, clarity)

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2. User System

- Implemented user registration and login functionality
- Defined two types of users:
  - Customer (User)
  - Staff (Admin)

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3. Food Ordering System

- The user can:
  - Select food items
  - Add items to the cart
  - View total price (in OMR)
  - Place an order

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4. Queue System

- Each order is initially set to:
  - Queued
- The system displays the user's position in the queue

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5. Order Status System

Implemented full order lifecycle:

- Queued
- Preparing
- Ready
- Completed

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6. Order Tracking

- Developed an order tracking page
- Displays order stages (Preparing → Cooking → Ready)
- Added animation (chef GIF)
- Notification and sound alert when the order is ready

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7. Rating System

- Users can rate orders using stars
- Ratings are stored in the database
- Ratings appear in the order history

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8. Order History

- Displays previous orders for each user
- Shows:
    - Food item
    - Price
    - Status
    - Rating

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9. Staff Panel

- View all orders
- View order status
- Identify the most ordered item

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10. Data Analysis

Added the following features:

- Chart (number of orders per item)
- Top Item
- beak Hour (most active ordering time)
- ⭐ Average Rating per item

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11. Database (SQLite)

- Created users table
- Created orders table
- Stored:
    - Orders
    - Status
    - Ratings
    - Time

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7. Testing Strategy and Evidence

7.1 Testing Strategy

The system was tested using:

- Unit testing for individual functions
- Functional testing for system features
- Manual testing for user interface interactions

7.2 Testing Evidence

Test Case	Description	Expected Result	Actual Result
Login	Valid credentials	Login successful	Passed
Register	New user	Account created	Passed
Order	Place order	Order saved in DB	Passed
Tracking	Track order	Status updates	Passed
Rating	Submit rating	Rating stored	Passed

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8. Security, Legal, Ethical, and Professional Considerations

- Security was considered during system development to protect user data and ensure safe interactions.
- User passwords are not stored in plain text; instead, hashing (SHA-256) is applied to enhance security.
- Input validation is considered to prevent invalid data entry and potential misuse.
- From a legal and ethical perspective, the system respects user privacy by limiting data collection to only necessary information.
- Professionally, the system was developed following structured coding practices, ensuring readability, maintainability, and scalability.
- These considerations ensure that the system aligns with best practices in software development

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10. Group Roles and Contributions

Member	Role	Contribution
Juhaina	Developer	**Implemented login**
Muna	Developer	**Developed UI and menu**
Amani	Analys	**Designed system and documentation**
Anhal	Analys	**Designed system and documentation**

11. Repository Link

GitHub Repository:

👉 (https://github.com/Software-Devlopment-Group/zahib-app/tree/main)